

Experiment 22: Bird Can Fly or Not Using Prolog

Aim

To develop a **Prolog program to determine whether a bird can fly or not** using facts, rules, and logical queries.

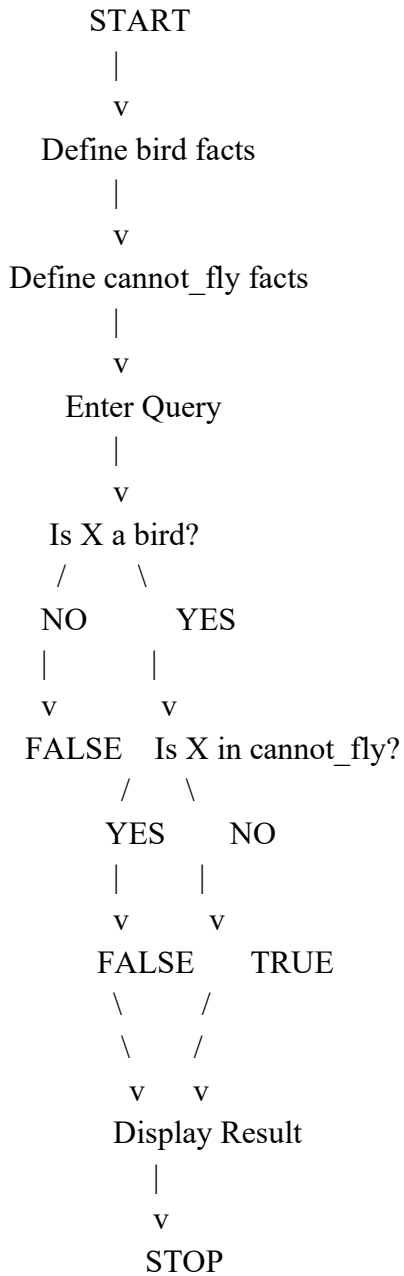
Objective

1. To represent birds and their flying abilities using Prolog facts.
2. To use rules for determining whether a bird can fly.
3. To handle exceptions such as penguins and ostriches that cannot fly.
4. To execute suitable queries and obtain the required results.

Algorithm

1. Start the program.
2. Define facts for different birds.
3. Define facts for birds that cannot fly.
4. Create a rule `can_fly(X)` to determine whether a bird can fly.
5. Check whether the given object is a bird.
6. Check whether the bird is not present in the `cannot_fly` facts.
7. If both conditions are satisfied, conclude that the bird can fly.
8. Otherwise, conclude that the bird cannot fly.
9. Execute the required queries.
10. Display the results.
11. Stop.

Flowchart



Prolog Code

```
% Bird facts
```

```
bird(sparrow).  
bird(eagle).  
bird(pigeon).  
bird(penguin).  
bird(ostrich).
```

```
% Birds that cannot fly
```

```
cannot_fly(penguin).  
cannot_fly(ostrich).
```

```
% Rule to determine whether a bird can fly
```

```
can_fly(X) :-  
    bird(X),  
    \+ cannot_fly(X).
```

Required Queries and Sample Output

Query 1: Check whether sparrow can fly

```
?- can_fly(sparrow).
```

Query 2: Check whether penguin can fly

```
?- can_fly(penguin).
```

Query 3: Check whether ostrich can fly

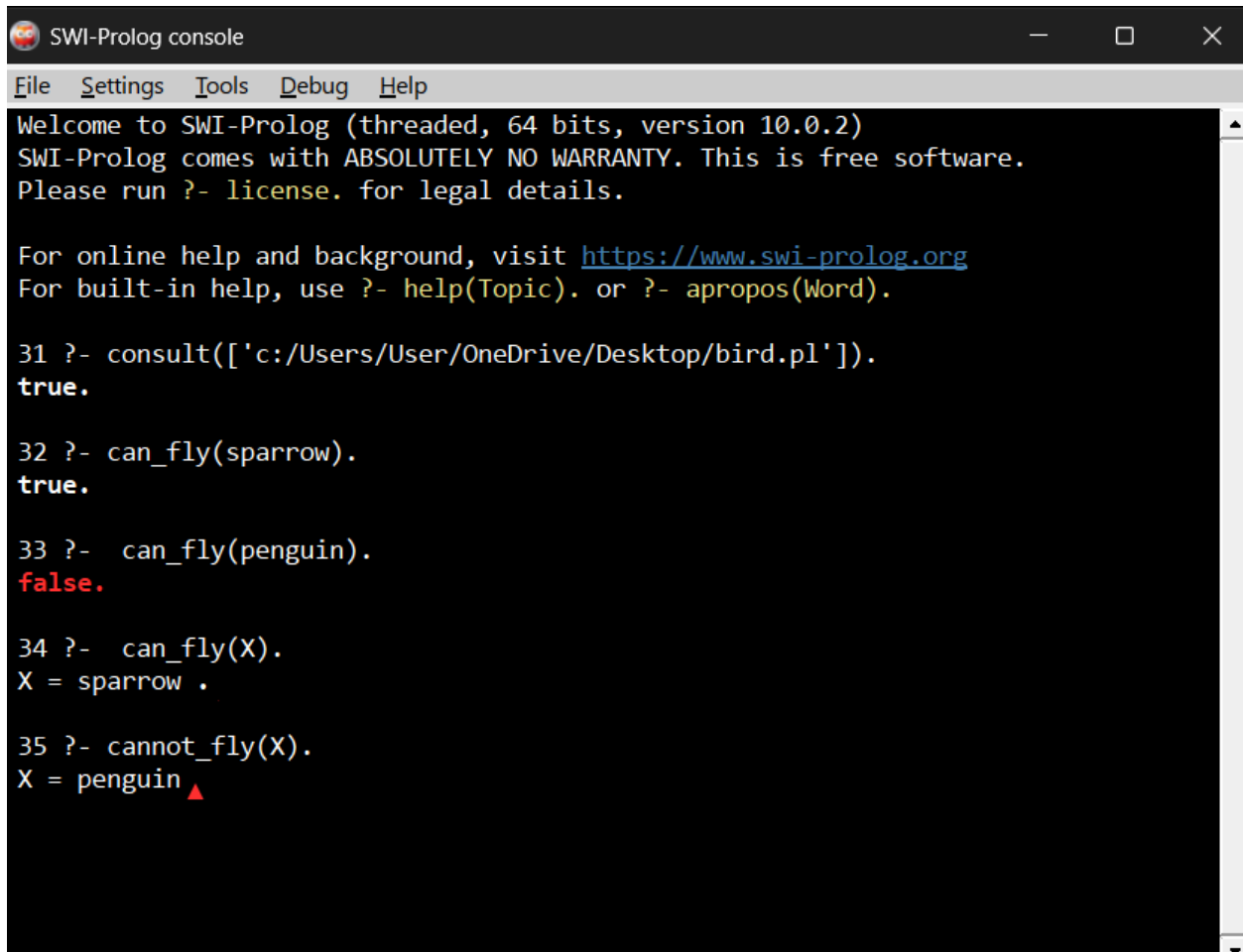
```
?- can_fly(ostrich).
```

Query 4: Find all birds that can fly

```
?- can_fly(X).
```

Query 5: Find all birds that cannot fly

?- cannot_fly(X).

A screenshot of the SWI-Prolog console window. The window has a title bar "SWI-Prolog console" and a menu bar with "File", "Settings", "Tools", "Debug", and "Help". The console text is as follows:

```
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

31 ?- consult(['c:/Users/User/OneDrive/Desktop/bird.pl']).
true.

32 ?- can_fly(sparrow).
true.

33 ?- can_fly(penguin).
false.

34 ?- can_fly(X).
X = sparrow .

35 ?- cannot_fly(X).
X = penguin ▲
```

Result

The Prolog program was successfully implemented to determine whether a given bird can fly or not. The required queries produced the expected results, correctly identifying flying and non-flying birds.

Conclusion

Thus, the experiment demonstrates the use of **Prolog facts, rules, negation, and queries** to solve a simple knowledge-based problem. The program successfully determines the flying ability of birds while handling exceptions such as penguins and ostriches.